FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XV

JEE (Main)-2021

TEST DATE: 18-07-2021

ANSWERS, HINTS & SOLUTIONS

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – A

SECTION – A

1. D

Sol. $B_2 - B_1 = \frac{2\pi mv}{q\ell}$

$(B_2 - B_1)^2 = \left(\frac{2\pi m}{q\ell}\right)^2 \left(\frac{2qV}{m}\right)$

2. B

Sol. shift = $(\ell - m)\left[1 - \frac{1}{n_1}\right] + m\left[1 - \frac{1}{n_2}\right]$

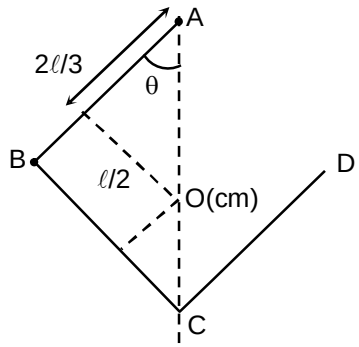
Shift = 0

3. B

Sol. $\tan\theta = \frac{OP}{AP} = \frac{\ell/2}{2\ell/3}$

$\tan\theta = \frac{3}{4}$

$\theta = \tan^{-1}\left(\frac{3}{4}\right)$



4. C

Sol. $F = \lambda v^2 + \lambda gy$

Total work done by the applied force = $\int_0^{\ell} F dy = \int_0^{\ell} (\lambda v^2 + \lambda gy)\, dy$

$\Rightarrow \quad \lambda v^2 \ell + \frac{\lambda g \ell^2}{2}$

Average power delivered = $\frac{\lambda v^2 \ell + \frac{\lambda g \ell^2}{2}}{\frac{\ell}{v}} = \lambda v^3 + \frac{\lambda \ell v g}{2}$

5. B

Sol. Upward force by capillary tube on top surface of liquid is f up = 4σ a $\cos\theta$

If liquid is raised to a height h then we use $4\sigma a\cos\theta = ha^2\rho g$ or $h = \frac{4\sigma\cos\theta}{a\rho g}$

6. A

Sol. Torricelli's formula

$v = \sqrt{2gy}$

According to equation of continuity

$Av_1 = av$

$A = \pi x^2$

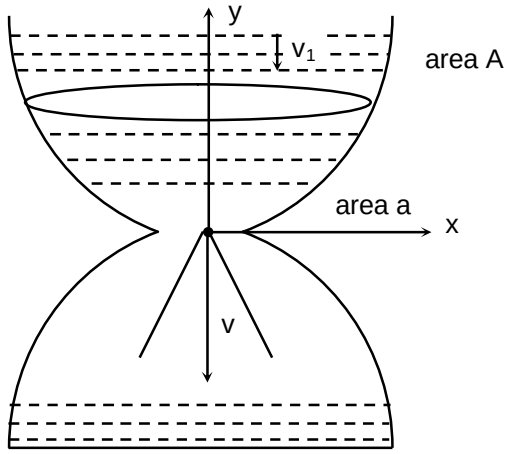
$\pi x^2 v_1 = a\sqrt{2gy}$

$$\frac{\pi x^2}{\sqrt{2gy}} = \frac{a}{v_1} = \text{conet} = \text{time}$$

$$\frac{\left(\pi x^2\right)^2}{2gy} = \frac{a^2}{v_1^2}$$

$$y = \left(\frac{\pi^2 v_1^2}{2ga^2}\right)x^4 = kx^4$$

Where $K = \dfrac{\pi^2 v_1^2}{2ga^2}$



7. B

Sol. Since frequency of first overtone are equal

$$\frac{3}{4\ell_1}\sqrt{\frac{B}{\rho_1}} = \frac{1}{\ell_2}\sqrt{\frac{B}{\rho_2}}$$

$$\ell_2 = \frac{4}{3}\ell_1\sqrt{\frac{\rho_1}{\rho_2}}$$

8. B

Sol. Optical path length = $\int_0^t \mu dx = \int_0^t \left(1+\frac{\alpha x}{t}\right)dx = \left(1+\frac{\alpha}{2}\right)t$

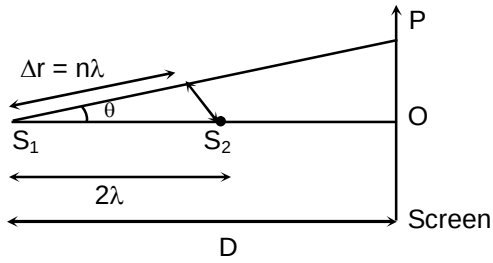
9. C

Sol. As $\cos\theta = \dfrac{n\lambda}{2\lambda} = \dfrac{1}{2}$ (for n = 1)

$\theta = 60°$.

$$\tan\theta = \frac{PO}{D}$$

$PO = D\sqrt{3}$



10. B

Sol. We know that $Z = \dfrac{E_0}{I_0}$

Given, $E_0 = 220$ and $I_0 = 10$

So $Z = \dfrac{220}{10} = 22$ Ohm

$$\phi = \frac{\pi}{6} - \left(-\frac{\pi}{6}\right) = \frac{\pi}{3}$$

$$P_0 = \frac{E_0}{\sqrt{2}}\frac{I_0}{\sqrt{2}}\cos\phi = \frac{220}{\sqrt{2}}\frac{10}{\sqrt{2}}\cos\frac{\pi}{3} = 550\ W$$

11. B

Sol. $i(t) = i_0 + i_1 e^{-t/RC} + i_2 e^{-t/2RC}$

where, $i_0 = i_1 = i_2 = \frac{E}{R}$

12. C

Sol. After collision A stops translation.

13. A

Sol. Area under p–x graph

$$\int P dx = \int \left(\frac{mdv}{dt}\right) v\, dx$$

$$\frac{1}{2}(2+4)10 = \int_{v_i}^{v_f} mv^2 dv$$

14. A

Sol. Loss in gravitational potential energy = Gain in spring potential energy

15. D

Sol. Due to variation in magnetic field there will be an induced current. Due to short circuiting bulb B goes out and resistances of circuit decreases hence current in A increases.

16. C

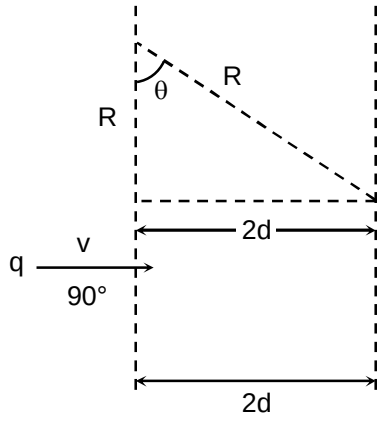
Sol. When an incident ray $\vec{I}$ is reflected by a mirror whose normal is $\vec{N}$, the reflected ray is given by:

$\vec{R} = \frac{-2(\vec{I}.\vec{N})\vec{N}}{\vec{N}.\vec{N}} + \vec{I}$

17. B

Sol. $t = \frac{2\pi m}{qB}\frac{\theta}{2\pi}$

$\sin\theta = \frac{2d}{R}$



18. A

Sol. $r + r' = A$

$r = A - r'$

$r = 60° - (10 + t^2) = 50 - t^2$

19. C

Sol. Time period of a spring block system

20. D

Sol. $S = L/2 + L/4 + L/6 + L/8 + L/10 = \dfrac{137L}{120}$

SECTION – B

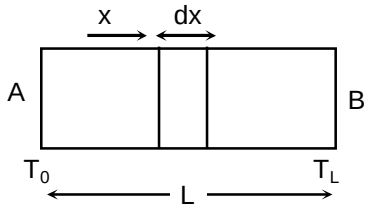
21. 00023.80

Sol. Energy released in the process = difference of energies of alpha particles and deuterons.

22. 00000.60

Sol. $v = \sqrt{\dfrac{\gamma RT}{M_0}} = \sqrt{\dfrac{5RT}{3M_0}}$

$$dx = vdt = \sqrt{\frac{5R}{3M_0}\left(T_0 + \frac{(T_L - T_0)x}{L}\right)}dt$$



23. 00000.50

Sol. $I_R = \dfrac{E_0}{R} = \dfrac{I_R}{\sqrt{2}} = \dfrac{E_0}{\sqrt{2}R} = \dfrac{E_0}{\sqrt{R^2 + \left(\omega L - \dfrac{1}{\omega C}\right)^2}}$

$$\Rightarrow \quad \omega_1 L - \frac{1}{\omega_1 C} = -R$$

$$\Rightarrow \quad \omega_2 L - \frac{1}{\omega_2 C} = +R$$

$$L(\omega_1 + \omega_2) = \left(\frac{\omega_1 + \omega_2}{\omega_1\omega_2}\right)\frac{1}{C} \Rightarrow \omega_1\omega_2 = \frac{1}{LC}$$

$$L(\omega_2 - \omega_1) + \left(\frac{\omega_2 - \omega_1}{\omega_1\omega_2}\right)\frac{1}{C} = 2R$$

24. 00007.00

Sol. U + 2 + 3nRT

$dU = n(3R)dT = nC_V dT \quad C_V = 3R$

25. 00002.00

Sol. $TV^{-1/2}$ = Constants

$\left(\frac{dv}{dT}\right)_{process} = \frac{2V}{T}$

$$C_P = C_V + \frac{P}{n}\left(\frac{dv}{dT}\right)$$

$$C_V + \frac{P}{n}\left[\frac{2V}{T}\right] = C_V + \frac{2PV}{nT} = C_V + 2R$$

26. 00008.00

Sol. Conserving angular momentum of the package about the centre of the planet and conserving mechanical energy.

27. 00204.80

Sol. For first resonance $\ell_1 + e = \lambda/4$

For second resonance $\ell_2 + e = 3\lambda/4$

and $\Delta v = 2f(\Delta\ell_2 + \Delta\ell_1)$

28. 00000.12

Sol. Tension at the lower end is 2g
Tension at the upper end is 8g
$V_2/V_1 = \sqrt{(T_2/T_1)}$
Since the frequency is constant
$V_2/V_1 = \lambda_2/\lambda_1 = 2$

29. 00000.50

Sol. $$\frac{KA(800-100)}{x} = \frac{7KA(800-0)}{\ell - x}$$

$$x = \frac{\ell}{9} = \frac{4.5}{9} = 0.50 \text{ m}$$

30. 00002.40

Sol. $\Delta y = (2\lambda D)/a = 2.40$ mm

Chemistry

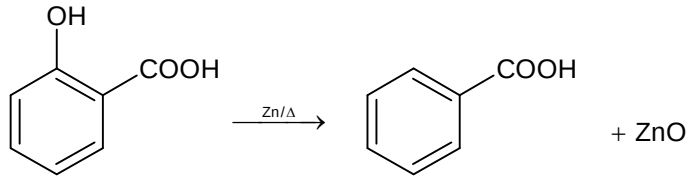
PART – B

SECTION – A

31. B

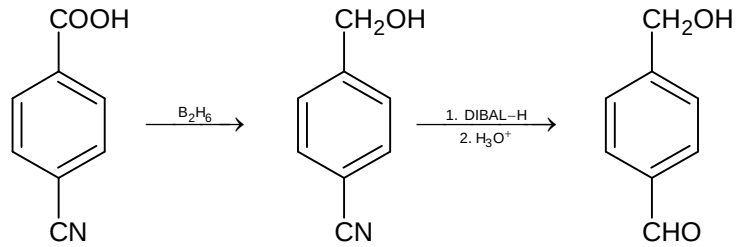
32. D

Sol.



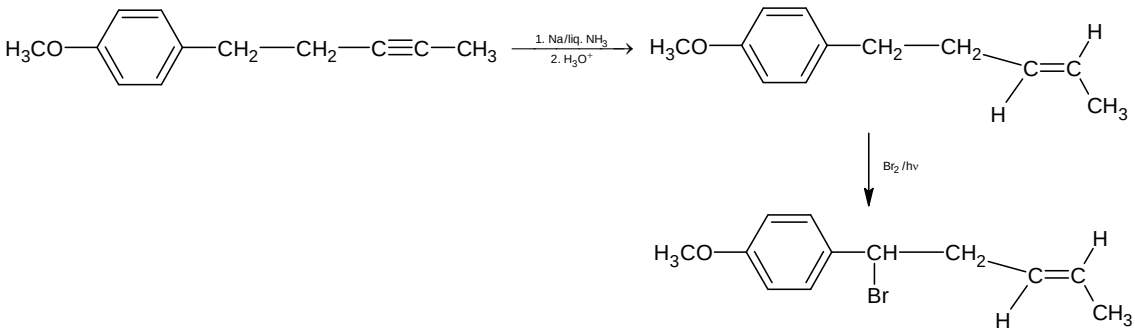
33. C

Sol.



34. C

Sol.



35. D

Sol. β – naphthol has phenolic group.

36. B

Sol. Moles of complex $= \frac{0.5}{266.5}$

$\because$ 1 moles gives = x moles of acid on cation exchange

$\therefore \frac{0.5}{266.5}$ moles gives = $\frac{x \times 1}{266.5} \times 0.5$

Hence, $\frac{x \times 1}{266.5} \times 0.5 = \frac{30.02 \times 0.125}{1000}$

$x = \frac{30.02 \times 0.125 \times 266.5}{0.5 \times 1000}$

x = 2

37. D

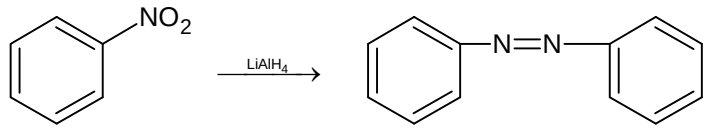
38. B

39. C

Sol. On moving down the group IP decreases.

40. B

Sol.



41. A

42. D

43. C

44. B

Sol. $N(CH_3)_3$ has tetrahedral structure.

45. D

46. D

47. B

Sol. gm eq of H_2O_2 = gm eq of $Na_2S_2O_3$

$N \times 25 = 0.3 \times 20$

$n = \frac{0.3 \times 20}{25} = 0.24$

$V = 5.6 \times 0.24 = 1.344$

48. A

49. D

50. D

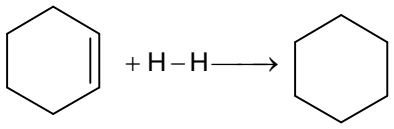
SECTION – B

51. 00002.00

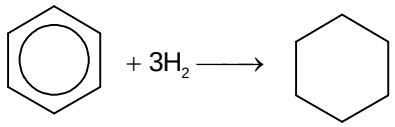
Sol. Conc. of $CuSO_4$ would not change hence $M = \frac{1}{500} \times 1000 = 2\ M$.

52. –00208.10

Sol.



$\Delta H = -119.5\ kJ\ mol^{-1}$



$\Delta H = ?$

$\left(\Delta_{hyd}H\right)_{theo} = \left(\Delta_hH\right)_{actual} + \left(\Delta H\right)_{resonance}$

$3x - 119.5 = \left(\Delta_hH\right)_{actual} + \left(-150.4\right)$

$\left(\Delta_hH\right)_{actual} = -358.5 + 150.4$

$= -208.1$

53. 00002.00

Sol. $\ell n\dfrac{k_2}{k_1} = \ell n\dfrac{A_2}{A_1} + \left(\dfrac{E_{a_1} - E_{a_2}}{RT}\right)$

$\ell n\dfrac{A_2}{A_1} = \dfrac{E_{a_2} - E_{a_1}}{RT} = \dfrac{2RT + E_{a_1} - E_{a_1}}{RT} = 2$

54. 00003.33

Sol.

$$A(g) + 2B(g) \rightleftharpoons 2C(g)$$

At eq. $\dfrac{4-x}{10}$ $\dfrac{4-2x}{10}$ $\dfrac{2x}{10}$ $2x = 2$

At eq. $\dfrac{3}{10}$ $\dfrac{2}{10}$ $\dfrac{2}{10}$ $x = 1$

$$K_c = \frac{\left(\frac{2}{10}\right)^2}{\left(\frac{3}{10}\right)\times\left(\frac{2}{10}\right)^2} = \frac{10}{3} = 3.33$$

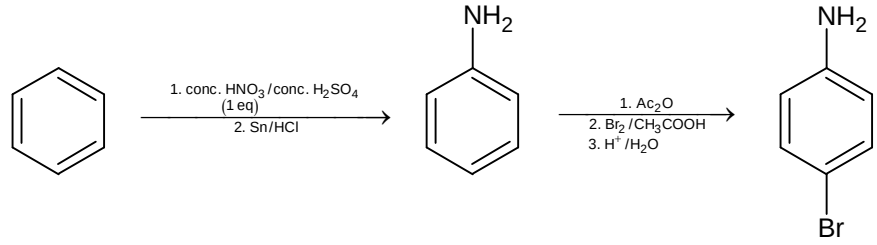
55. 00004.00

Sol. $\dfrac{P_A}{P_T} = \dfrac{n_A}{n_A + n_B}$

$\dfrac{4}{6} = \dfrac{\frac{1}{M_A}}{\frac{1}{M_A} + \frac{2}{M_B}}$, $\dfrac{M_B}{M_A} = 4$

56. 00001.00

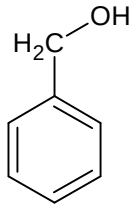
Sol.



57. 00014.81

Sol

$$CH_3(CH_2)_5CH_3 \xrightarrow[\substack{2.\ Br_2/h\nu \\ 3.\ Mg/ether \\ 4.\ H_2O}]{1.\ Cr_2O_3/Al_2O_3,\ 773K,\ 10-20\ atm}$$



58. 00008.00

Sol. n = 3 there is no symmetry hence total stereoisomers $= 2^n = 2^3 = 8$.

59. 00001.50

Sol. $O_2^- = \sigma_{1s}^2, \sigma_{1s}^{*2}\sigma_{2s}^2, \sigma_{2s}^{*2}, \sigma_{2pz}^2, \sigma_{2px}^2 = \sigma_{2py}^2, \sigma_{2px}^{*2} = \sigma_{2py}^{*1}$

Bond order $= \dfrac{6-3}{2} = 1.5$

60. 00000.50

Sol. $6XeF_4 + 12H_2O \longrightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$

Mathematics

PART – C

SECTION – A

61. A

Sol. For any $a \in N$, we find that a|a, therefore R is reflexive but R is not symmetric, because aRb does not imply that bRa.

62. A

Sol. Use Expansion

63. D

Sol. Given that $z^2 + (p + iq)z + r + is = 0$ ..... (1)

Let $z = \alpha$ (where α is real) be a root of (1), then

$\alpha^2 + (p + iq)\alpha + r + is = 0$

or $\alpha^2 + p\alpha + r + i(q\alpha + s) = 0$

Equating real and imaginary parts, we have $\alpha^2 + p\alpha + r = 0$ and $q\alpha + s = 0$

Eliminating α, we get $\left(\frac{-s}{q}\right)^2 + p\left(\frac{-s}{q}\right) + r = 0$

or $s^2 - pqs + q^2r = 0$ or $pqs = s^2 + q^2r$

64. A

Sol. n^{th} term of the series is $20 + (n-1)\left(-\frac{2}{3}\right)$

For sum to be maximum, n^{th} term ≥ 0

$\Rightarrow 20 + (n-1)\left(-\frac{2}{3}\right) \geq 0 \Rightarrow n \leq 31$

Thus the sum of 31 terms is maximum and is equal to

$\frac{31}{2}\left[40 + 30 \times \left(-\frac{2}{3}\right)\right] = 310$

65. D

Sol. If the roots of the quadratic equation $ax^2 + bx + c = 0$ exceed a number k, then $ak^2 + bk + c > 0$ if $a > 0$, $b^2 - 4ac \geq 0$ and sum of the roots $> 2k$

Therefore, if the roots of $x^2 + x + a = 0$ exceed a number a, then

$a^2 + a + a > 0$, $1 - 4a \geq 0$ and $-1 > 2a$

$\Rightarrow a(a+2) > 0$, $a \leq \frac{1}{4}$ and $a < -\frac{1}{2}$

$\Rightarrow a > 0$ or $a < -2$, $a < \frac{1}{4}$ and $a < -\frac{1}{2}$

Hence, $a < -2$

66. B

Sol. Since the number of students giving wrong answers to at least i question $(i = 1, 2, ..., n) = 2^{n-i}$

The number of students answering exactly i $(1 \leq i \leq -1)$ questions wrongly

= {the number of students answering at least i questions wrongly, $i = 1, 2, ...,n)$}

– {the number of students answering at least $(i+1)$ questions wrongly

$(2 \le i+1 \le n)\} = 2^{n-i} - 2^{n-(i+1)} (1 \le i \le n-1)$

Now, the number of students answering all the n questions wrongly $= 2^{n-n} = 2^0$

Thus the total number of wrong answers

$= 1(2^{n-1} - 2^{n-2} + 2(2^{n-2} - 2^{n-3}) + 3(2^{n-3} - 2^{n-4}) + ... + (n-1)(2^1 - 2^0) + n(2^0)$

$= 2^{n-1} + 2^{n-2} + 2^{n-3} + ... + 2^0 = 2^n - 1$ ($\because$ Its a G.P.)

67. D

Sol. $(1+x+x^3+x^4)^{10} = (1+x)^{10}(1+x^3)^{10}$

$= (1+ {}^{10}C_1.x + {}^{10}C_2.x^2 + ...)(1+ {}^{10}C_1.x^3 + {}^{10}C_2.x^6 + ...)$

$\therefore$ Coefficient of $x^4 = {}^{10}C_1.{}^{10}C_1 + {}^{10}C_4 = 310$

68. D

Sol. $\begin{vmatrix} x+a & b & c \\ b & x+c & a \\ c & a & x+b \end{vmatrix} = 0$

$\Rightarrow (x+a+b+c)\begin{vmatrix} 1 & b & c \\ 1 & x+c & a \\ 1 & a & x+b \end{vmatrix} = 0$

$\Rightarrow x = -(a+b+c)$ is one of the root of the equation

69. B

Sol. $x\sin\varphi = \tan\theta - x\cos\varphi\tan\theta$

$\Rightarrow x = \dfrac{\tan\theta}{\sin\varphi + \cos\varphi\tan\theta} = \dfrac{\sin\theta}{\cos\theta\sin\varphi + \cos\varphi\sin\theta} = \dfrac{\sin\theta}{\sin(\theta+\varphi)}$

Similarly, $y = \dfrac{\sin\varphi}{\sin(\theta+\varphi)}$; $\therefore \dfrac{x}{y} = \dfrac{\sin\theta}{\sin\varphi}$

70. A

Sol. On expanding determinant,

$\cos^2(A+B) + \sin^2(A+B) + \cos 2B = 0$

$1 + \cos 2B = 0$ or $\cos 2B = \cos\pi$

or $2B = 2n\pi + \pi$ or $B = (2n+1)\dfrac{\pi}{2}, n \in Z.$

71. A

Sol. By the help of given condition of $a+b+c=0$, the three lines reduce to $x - y = \dfrac{p}{a}$ or $\dfrac{p}{b}$ or $\dfrac{p}{c}(p \ne 0)$.

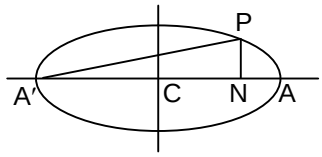
All these lines are parallel. Hence they do not intersect in finite plane

72. C

Sol. Required area $= \dfrac{a}{h^2+k^2}(h^2+k^2-a^2)^{3/2} = \dfrac{3}{4^2+3^2}(4^2+3^2-9)^{3/2} = \dfrac{192}{25}$

73. A

Sol. Let ellipse be $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$



$P=(a\cos\theta,b\sin\theta), A$ and $A'\equiv(\pm a,0), N\equiv(a\cos\theta,0),$

$PN=b\sin\theta,\ AN=a(1-\cos\theta),$

$A'N=a(1+\cos\theta)$

$$\frac{(PN)^2}{AN\,A'N}=\frac{b^2\sin^2\theta}{a^2(1-\cos\theta)(1+\cos\theta)}=\frac{b^2}{a^2}$$

74. A

Sol. The vector equation of a plane through the line of intersection of the planes $r.(i+3j-k)=0$ and $r.(j+2k)=0$ can be written as

$(r.(i+3j-k))+\lambda(r.(j+2k))=0$ ..... (1)

This passes through $2i+j-k$

$\therefore (2i+j-k).(i+3j-k)+\lambda(2i+j-k).(j+2k)=0$

or $(2+3+1)+\lambda(0+1-2)=0\Rightarrow\lambda=6$

Put the value of λ in (1), we get

$r.(i+9j+11k)=0$, which is the required plane

75. A

Sol. Force $F=\overrightarrow{AB}=(3-1)\mathbf{i}+(-4-2)\mathbf{j}+(2+3)\mathbf{k}=2\mathbf{i}-6\mathbf{j}+5\mathbf{k}$

Moment of Force $\vec{F}$ w.r.t $M=\overrightarrow{MA}\times\vec{F}$

$\because\ \overrightarrow{MA}=(1+2)\mathbf{i}+(2-4)\mathbf{j}+(-3+6)\mathbf{k}=3\mathbf{i}-2\mathbf{j}+3\mathbf{k}$

Now, $\overrightarrow{MA}\times\vec{F}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\3&-2&3\\2&-6&5\end{vmatrix}=\mathbf{i}(-10+18)+\mathbf{j}(6-15)+\mathbf{k}(-18+4)=8i-9j-14k$

76. A

Sol. $[xy\cos(xy)+\sin(xy)]dx+x^2\cos(xy)dy=0$

$xy\cos(xy)dx+x^2\cos(xy)dy+\sin(xy)dx=0$

$x\cos(xy)(ydx+xdy)+\sin(xy)dx=0$

$\cot(xy)dxy+\frac{dx}{x}=0$

$\log\sin(xy)+\log x=k\Rightarrow x\sin(xy)=k$

77. A

Sol. $ye^{-x/y}dx-(xe^{-x/y}+y^3)dy=0$

$e^{-x/y}(ydx-xdy)=y^3dy\ \Rightarrow\ e^{-x/y}\frac{(ydx-xdy)}{y^2}=ydy$

$e^{-x/y}d\left(\frac{x}{y}\right)=ydy$. Integrating both sides, we get $k-e^{-x/y}=\frac{y^2}{2}\Rightarrow\frac{y^2}{2}+e^{-x/y}=k$

78. C

Sol. Let A be the event that face 4 turns up and B be the event that face 5 turns up then $P(A) = 0.25, P(B) = 0.05$

Since A and B are mutually exclusive, so $P(A \cup B) = P(A) + P(B) = 0.25 + 0.05 = 0.30$

We have to find $P\left(\frac{A}{A \cup B}\right)$, which is equal to $P\frac{[A \cap (A \cup B)]}{P(A \cup B)} = \frac{P(A)}{P(A \cup B)} = \frac{0.25}{0.30} = \frac{5}{6}$

79. A

Sol. $\sim [p \vee (\sim p \vee q)] \equiv \sim p \wedge \sim (\sim p \vee q)$

$\equiv \sim p \wedge (\sim (\sim p) \wedge \sim q)$

$\equiv \sim p \wedge (p \wedge \sim q)$

80. B

Sol. Tangent at $(a\sec\theta, b\tan\theta)$ is, $\frac{x}{(a/\sec\theta)} - \frac{y}{(b/\tan\theta)} = 1$ or $\frac{a}{\sec\theta} = 1, \frac{b}{\tan\theta} = 1$

$\Rightarrow a = \sec\theta$, $b = \tan\theta$ or (a,b) lies on $x^2 - y^2 = 1$

SECTION – B

81. 00003.00

Sol. $\sin^{-1}x = 2\tan^{-1}x \Rightarrow \sin^{-1}x = \sin^{-1}\frac{2x}{1+x^2}$

$\Rightarrow \frac{2x}{1+x^2} = x \Rightarrow x^3 - x = 0$

$\Rightarrow x(x+1)(x-1) = 0 \Rightarrow x = \{-1, 1, 0\}$

82. 00006.00

Sol. $\theta = \tan^{-1}\left(\frac{2\sqrt{h^2 - ab}}{a+b}\right) = \tan^{-1}\left(\frac{\sqrt{4h^2 - 4ab}}{a+b}\right) = \tan^{-1}\left(\frac{\sqrt{3a^2 + 3b^2 + 10ab - 4ab}}{a+b}\right) = 60^\circ$

83. 00000.00

Sol. $f(x) = \frac{\cos^2 x + \sin^4 x}{\sin^2 x + \cos^4 x} \Rightarrow f(x) = \frac{\cos^2 x + \sin^2 x(1 - \cos^2 x)}{\sin^2 x + \cos^2 x(1 - \sin^2 x)}$

$\Rightarrow f(x) = \frac{\sin^2 x + \cos^2 x - \sin^2 x\cos^2 x}{\sin^2 x + \cos^2 x - \sin^2 x\cos^2 x}$

$\Rightarrow f(x) = 1$

84. 00001.00

Sol. $\lim\limits_{x\to 0} \frac{x(1 + a\cos x) - b\sin x}{x^3} = 1$

$\Rightarrow \lim\limits_{x\to 0} \frac{x\left\{1 + a\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots\right)\right\} - b\left\{x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots\right\}}{x^3} = 1$

$$\Rightarrow \lim_{x\to 0}\frac{(1+a-b)+x^2\left(\frac{b}{3!}-\frac{a}{2!}\right)+x^4\left(\frac{a}{4!}-\frac{b}{5!}\right)+...}{x^2}=1 \quad ..... (1)$$

If $1+a-b\neq 0$, then L.H.S. $\to\infty$ as $x\to 0$ while R.H.S. = 1, therefore $1+a-b=0$

Now from equation (1), $\lim_{x\to 0}\frac{x^2\left(\frac{b}{3!}-\frac{a}{2!}\right)+x^4\left(\frac{a}{4!}-\frac{b}{5!}\right)+...}{x^2}=1$

$\Rightarrow \frac{b}{3!}-\frac{a}{2!}=1 \Rightarrow b-3a=6$

Solving $1+a-b=0$ and $b-3a=6$, we get $a=-5/2, b=-3/2$

85. 00004.00

Sol. The function is monotonic increasing, if $f'(x)>0$

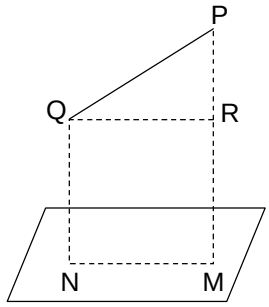
$$\Rightarrow \frac{(2\sin x+3\cos x)(\lambda\cos x-6\sin x)}{(2\sin x+3\cos x)^2}-\frac{(\lambda\sin x+6\cos x)(2\cos x-3\sin x)}{(2\sin x+3\cos x)^2}>0$$

$\Rightarrow 3\lambda(\sin^2 x+\cos^2 x)-12(\sin^2 x+\cos^2 x)>0$

$\Rightarrow 3\lambda-12>0 \Rightarrow \lambda>4$

86. 00001.41

Sol. Given plane is $x+y+z-3=0$. From point P and Q draw PM and QN perpendicular on the given plane and QR $\perp$MP



$$|MP|=\frac{0+1+0-3}{\sqrt{1^2+1^2+1^2}}=\frac{-2}{\sqrt{3}}, \quad |NQ|=\frac{-2}{\sqrt{3}}$$

$$|PQ|=\sqrt{(0-0)^2+(0-1)^2+(1-0)^2}=\sqrt{2}$$

$$|RP|=|MP|-|MR|=|MP|-|NQ|=0$$

$$\therefore |NM|=|QR|=\sqrt{PQ^2-RP^2}=\sqrt{(\sqrt{2})^2-0}=\sqrt{2}$$

87. 00001.00

Sol. $$I=\int\frac{dx}{\sin x-\cos x+\sqrt{2}}$$

$$=\int\frac{dx}{\sqrt{2}(\sin x.\sin\frac{\pi}{4}-\cos x\cos\frac{\pi}{4}+1)}$$

$$=\frac{1}{\sqrt{2}}\int\frac{dx}{1-\cos\left(x+\frac{\pi}{4}\right)}=\frac{1}{\sqrt{2}}\int\frac{dx}{1-\cos 2\left(\frac{x}{2}+\frac{\pi}{8}\right)}$$

$$=\frac{1}{\sqrt{2}}\int\frac{dx}{2\sin^2\left(\frac{x}{2}+\frac{\pi}{8}\right)}=\frac{1}{2\sqrt{2}}\int\text{cosec}^2\left(\frac{x}{2}+\frac{\pi}{8}\right)dx$$

$$=\frac{1}{2\sqrt{2}}\frac{-\cot\left(\frac{x}{2}+\frac{\pi}{8}\right)}{1/2}+c=\frac{-1}{\sqrt{2}}\cot\left(\frac{x}{2}+\frac{\pi}{8}\right)+c$$

88. 00004.00

Sol. $I = \int_0^{\infty} \frac{xdx}{(1+x)(1+x^2)}$

Put $x = \tan\theta$, we get

$I = \int_0^{\pi/2} \frac{\tan\theta}{1+\tan\theta} d\theta = \int_0^{\pi/2} \frac{\sin\theta}{\cos\theta+\sin\theta} d\theta = \frac{\pi}{4}$

89. 00012.56

Sol. Required area $= \int_0^{\pi} y\, dx + \left|\int_{\pi}^{2\pi} y\, dx\right| = 4\pi$ sq.unit

90. 00009.00

Sol. $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT$

$n(E) = 4, n(F) = 4$ and $n(E \cap F) = 3$